

Blue/Green Deployment on AWS

A simple, step-by-step guide for everyone

Release a new app version with zero downtime using an Application Load Balancer

What you will build

Two copies of a small website (an old one called **Blue** and a new one called **Green**) running on AWS, with a load balancer in front. You will then slowly move visitors from Blue to Green, without the website ever going down.

Platform	AWS (KodeKloud Labs), Region us-east-1
Services used	VPC, EC2, Target Groups, Application Load Balancer
Difficulty	Beginner (no coding needed, just copy and paste)
Time needed	About 45 to 60 minutes
Date	07 October 2026

How to read this guide

This guide explains every step in plain words. If you see a technical word, check the **Simple Dictionary** on the next page.

1. The Big Idea in Plain Words

Imagine a restaurant that wants to change its menu. Instead of closing for a day, it opens a **second kitchen** with the new menu. At first, only 1 in 5 customers is sent to the new kitchen. If they like it, everyone is sent there. If something goes wrong, customers are sent back to the old kitchen in seconds.

That is exactly what we do with software:

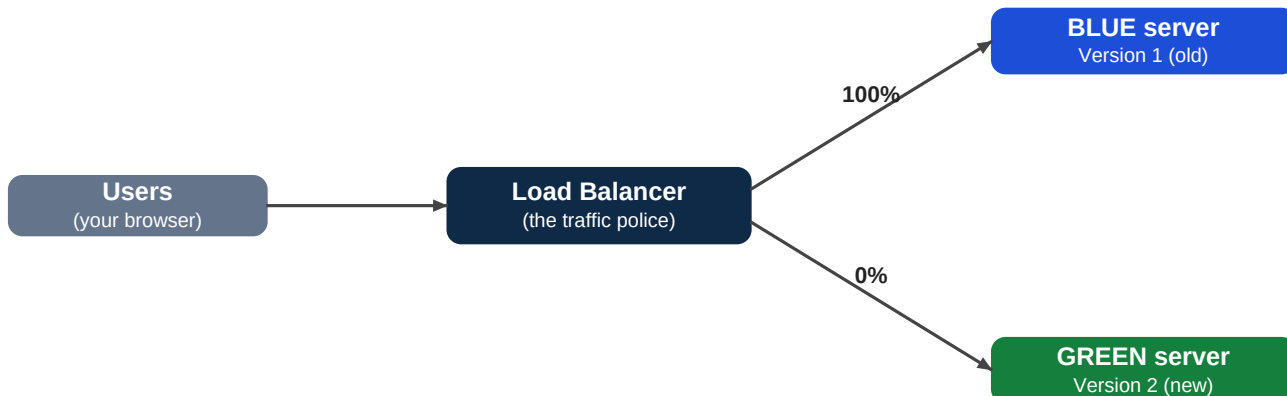


Figure 1: At the start, 100% of users go to Blue.

Why this is useful

- **No downtime:** the old version keeps running while the new one is tested.
- **Low risk:** only a small share of users (20%) see the new version first. This is called a **canary release**.
- **Easy rollback:** if Green has problems, set Blue back to 100%.
- **Full cutover:** when Green is healthy, send 100% of users to it.

Simple Dictionary

Word	What it means (simple)
AWS	Amazon's cloud. You rent computers and network tools by the hour.
VPC	Your own private network inside AWS, like a fenced plot of land.
Subnet	A smaller section of the VPC. A <i>public</i> subnet can reach the internet.
Availability Zone (AZ)	A separate AWS data centre (like 1a, 1b). Using two protects you if one fails.
Internet Gateway	The front gate that connects your VPC to the internet.
Route table	A road map telling traffic where to go.
Security group	A firewall that decides who can come in (here: anyone on port 80).
EC2	A virtual computer (server) in AWS.
User data	A script that runs by itself when the server starts for the first time.
Target group	A list of servers that the load balancer can send visitors to.
Load balancer (ALB)	The traffic police. One address for users, it decides which server answers.

Word	What it means (simple)
Canary deployment	Releasing a new version to a few users first, to check it is safe.

2. What We Are Building (Architecture)

Everything below is created in the AWS Console. Names are written exactly as used in this project so you can follow along.

Component	Name	Details
VPC	BG-VPC	10.0.0.0/16
Public subnet 1	Public-1	us-east-1a, 10.0.1.0/24
Public subnet 2	Public-2	us-east-1b, 10.0.2.0/24
Internet Gateway	BG-IGW	Attached to BG-VPC
Route table	BG-RT	0.0.0.0/0 to BG-IGW; both subnets linked
Security group	Web-SG	Allows HTTP port 80 from anywhere
EC2 (Blue)	Blue-Server	t3.micro, Amazon Linux 2023, in Public-1
EC2 (Green)	Green-Server	t3.micro, Amazon Linux 2023, in Public-2
Target groups	TG-Blue, TG-Green	HTTP 80, one server each
Load balancer	BlueGreen-ALB	Internet-facing, listener HTTP:80

Traffic flow: User > ALB (HTTP:80) > weighted rule > TG-Blue (Blue-Server) or TG-Green (Green-Server).

Why two subnets?

An ALB must live in at least **two subnets in different Availability Zones**. That is why we build two public subnets first.

Quick checklist before you start

- An AWS account or a KodeKloud AWS lab with Region **us-east-1** selected (top-right of the console).
- A browser. Use an **incognito window** for testing so cached pages do not confuse you.
- A notepad to paste the two small scripts used later.

3. Task 1: Build the Network

Goal: a private network with two public subnets that can reach the internet.

Step 1: Create the VPC

- Open the **VPC** service and click **Create VPC**.
- Choose **VPC only**. Name: **BG-VPC**. IPv4 CIDR: **10.0.0.0/16**.
- Click **Create VPC**.

Step 2: Create two public subnets

Name	Availability Zone	CIDR
Public-1	us-east-1a	10.0.1.0/24
Public-2	us-east-1b	10.0.2.0/24

Step 3: Turn on auto-assign public IP

- Select **Public-1**, then **Actions > Edit subnet settings**.
- Tick **Enable auto-assign public IPv4 address** and Save.
- Repeat for **Public-2**.

Why?

Without a public IP, your server cannot be reached from the internet.

Step 4: Internet Gateway and routes

- Create an Internet Gateway named **BG-IGW** and **attach** it to BG-VPC.
- Create a route table named **BG-RT** in BG-VPC.
- Add route **0.0.0.0/0** with target **BG-IGW** (this means: all internet traffic goes through the gate).
- Associate **Public-1** and **Public-2** with BG-RT.

Result

A custom VPC with two public subnets that have internet access.

4. Task 2: Blue Server (Version 1)

Goal: a server that shows a blue page. This is the current live version.

- Go to **EC2 > Launch instances**. Name: **Blue-Server**.
- AMI: **Amazon Linux 2023**. Type: **t3.micro**. Key pair: **none** (we do not need to log in).
- Network: VPC **BG-VPC**, subnet **Public-1**.
- Create a new security group **Web-SG** with inbound **HTTP (80)** from **0.0.0.0/0**.
- Open **Advanced details > User data**, paste the script below, then launch.

```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo '<html><body style="background-color:blue; color:white">
<h1>VERSION 1 (BLUE)</h1></body></html>' > /var/www/html/index.html
```

What does this script do?

It installs Apache (a web server), starts it, makes it start automatically after a reboot, and writes a small blue web page. *Note: in the real script the echo command is written on one line.*

5. Task 3: Green Server (Version 2)

Goal: a second server that shows a green page. This is the new version we want to test.

- **EC2 > Launch instances.** Name: **Green-Server**.
- Same AMI and type (Amazon Linux 2023, t3.micro), no key pair.
- Network: VPC **BG-VPC**, subnet **Public-2** (the other Availability Zone).
- Security group: **Select existing > Web-SG** (untick the default one).
- User data: the same script, but with the green page line:

```
echo '<html><body style="background-color:green; color:white">
<h1>VERSION 2 (GREEN)</h1></body></html>' > /var/www/html/index.html
```

Check before continuing

Both servers must be in **BG-VPC**, use **Web-SG**, and allow port 80. Wait until both show **2/2 checks passed** before moving on.

6. Task 4: Target Groups

Goal: tell the load balancer which servers belong to Blue and which to Green.

- **EC2 > Target Groups > Create target group.**
- Type: **Instances**. Name: **TG-Blue**. VPC: **BG-VPC**. Protocol: **HTTP:80**.
- Select **Blue-Server**, click **Include as pending below**, then create.
- Repeat with name **TG-Green** and register **Green-Server**.

Target group	Registered server
TG-Blue	Blue-Server
TG-Green	Green-Server

7. Task 5: Application Load Balancer

Goal: one single address for users that decides which target group answers.

- **EC2 > Load Balancers > Create load balancer > Application Load Balancer.**
- Name: **BlueGreen-ALB** (Internet-facing).
- Network mapping: VPC **BG-VPC**, tick **us-east-1a (Public-1)** and **us-east-1b (Public-2)**.
- Security group: remove the default, select **Web-SG**.
- Listener **HTTP:80**, default action: forward to **TG-Blue**.
- Create it and wait until the state is **Active** (a few minutes).

Now copy the **DNS name** of the load balancer and open it in the browser using **http://**. You should see:

VERSION 1 (BLUE)

Figure 2: The default rule forwards to TG-Blue, so Blue is shown.

8. Task 6: Shift the Traffic (Canary)

Goal: move users from Blue to Green slowly and safely.

Step 1: Check the starting point (100% Blue)

Open the ALB DNS name and refresh several times. The page should **always** be Blue.

Step 2: Send 20% of users to Green

- Open the Load Balancer > **Listeners and rules** > select **HTTP:80** > **Manage rules** > **Edit rule**.
- Under Routing actions set weights: **TG-Blue = 80** and **TG-Green = 20**.
- Click **Save changes**.

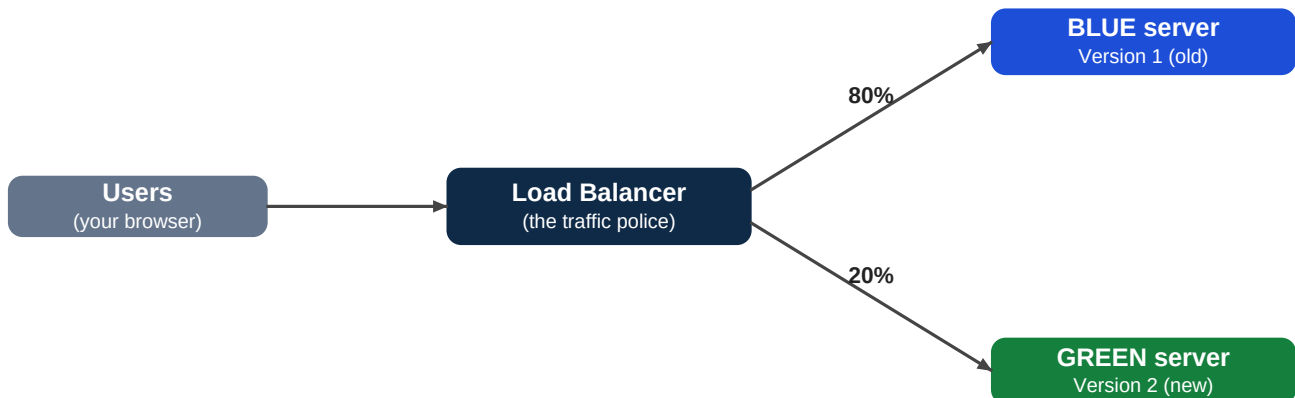


Figure 3: Canary stage. 8 out of 10 users still see Blue, 2 out of 10 see Green.

Step 3: Wait 30 to 60 seconds

The new rule needs a short time to apply.

Step 4: Test the shift

Refresh many times. Most of the time you will see Blue, and about 1 in 5 times you will see Green. That matches the 80/20 weights.

Proof from the test: a short screen recording was taken while refreshing the ALB address. These frames are taken from that video. Blue appears most of the time and Green shows up now and then, exactly what an 80/20 split should look like.



Figure 4: Frames from the test video. Most frames show VERSION 1 (BLUE); some show VERSION 2 (GREEN) from TG-Green. (The original video file is kept alongside this guide.)

Tip

Weighted routing is random, so with only a few refreshes you might see Green twice in a row or not at all. Refresh 20 to 30 times, or use a new incognito window.

Step 5: Full cutover (100% Green)

- Edit the rule again.
- Set weights: **TG-Blue = 0** and **TG-Green = 100**.
- Click **Save changes**.

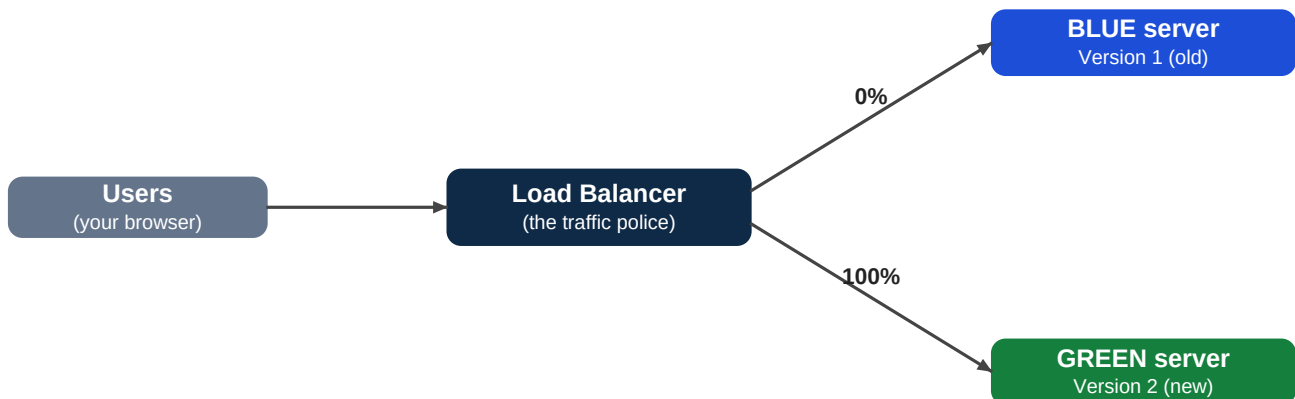


Figure 5: Full cutover. Every user now sees Green.

Result

All users now see VERSION 2 (GREEN). The Blue server is still running, so a rollback only needs the weights changed back.

9. Weight Summary (Cheat Sheet)

Stage	TG-Blue	TG-Green	User sees
Initial	100	0	Always Blue
Canary	80	20	Mostly Blue, sometimes Green
Full cutover	0	100	Always Green
Rollback (if needed)	100	0	Always Blue

10. Common Problems and Fixes

Problem	Likely reason and fix
Page does not open	Use http:// , not https . Check Web-SG allows port 80 and the ALB state is Active.
503 or 502 error	Targets are not healthy yet. Wait a few minutes and check the target group health status.
Targets show Unhealthy	User data may have failed. Check the script was pasted correctly and Apache is running.
Always shows the same colour	Browser cache. Use an incognito window and refresh many times.
Cannot create the ALB	Select two subnets in different Availability Zones.

11. Clean Up (Avoid Charges)

In a personal AWS account, running resources cost money. Delete them in this order when you finish:

- Delete the **load balancer** (BlueGreen-ALB).
- Delete the **target groups** (TG-Blue, TG-Green).
- **Terminate** both EC2 instances.
- Delete the security group, route table, internet gateway (detach first), subnets, then the VPC.

12. What You Learned

- **VPC networking:** multi-AZ public subnets, internet gateway and route tables.
- **EC2:** automatic web server setup with user data.
- **Target groups:** separating environments for traffic control.
- **Application Load Balancer:** listeners and weighted routing rules.
- **Canary deployment:** releasing a new version to a small share of users first.

13. Conclusion

This project shows how a new application version can be released with zero downtime. Two environments behind one load balancer, plus weighted rules, reduce risk and make rollback quick and simple. The same idea is used by real companies to release software safely.